

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

REVIEW — Parallel Resistors

Keep this sheet. *Parallel is the mirror image of series — what was shared is now equal, and what was equal is now shared.*

Key concept

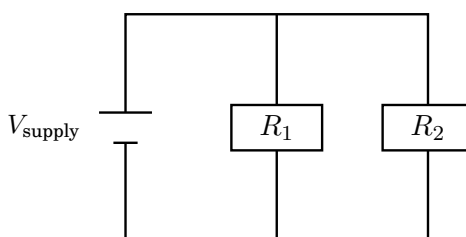
THE THREE RULES — parallel (branches, so the charge has a choice of route)

- Potential difference is the same across every branch.** $V_1 = V_2 = V_{\text{supply}}$
Each branch is connected straight across the supply, so each one gets the full p.d.
- Branch currents add up.** $I_{\text{total}} = I_1 + I_2$
The current splits at the junction and rejoins after it. Nothing is lost.
- Total resistance is less than the smallest single resistor.**
Adding a resistor in parallel always *decreases* total resistance — you have opened another route for the charge, so more current flows for the same p.d.

Mark scheme

You are NOT required to calculate the total resistance of two resistors in parallel.

(AQA Trilogy, aP2-21.) You *are* required to explain *qualitatively* why it goes down (aP2-22), and to use the two rules above. Calculations with $R_{\text{total}} = R_1 + R_2$ belong to **series** circuits only.



current splits at the junction

	Series	Parallel
Current	same everywhere	<i>splits</i> — branches add to the total
P.d.	<i>shared</i> between components	same across every branch
Total resistance	goes <i>up</i> $R_1 + R_2$	goes <i>down</i> — less than the smallest
Remove one	everything stops	the other branch keeps working

Worked example

WORKED EXAMPLE 1 — the standard one

A 4Ω and a 6Ω resistor are connected in parallel across a 12V supply. Find the current in each branch, and the total current.

1. P.d. on each branch Each branch gets the *full* 12V . (rule 1)

2. Branch 1 $I_1 = \frac{V}{R_1} = \frac{12}{4} = 3.0\text{A}$

3. Branch 2 $I_2 = \frac{V}{R_2} = \frac{12}{6} = 2.0\text{A}$

4. Total current $I_{\text{total}} = I_1 + I_2 = 3.0 + 2.0 = 5.0\text{A}$ (rule 2)

Notice: the smaller resistor takes the *bigger* current — the opposite of what happens to p.d. in series.

REVIEW — Parallel Resistors: worked example & practice

Worked example

WORKED EXAMPLE 2 — proving rule 3 without calculating it

Using the circuit above, show that the total resistance must be less than $4\ \Omega$.

Each resistor on its own would draw less current than the pair does together. With both branches connected, $5.0\ \text{A}$ flows out of the supply — more than the $3.0\ \text{A}$ that the $4\ \Omega$ resistor draws by itself. Same p.d., more current, so the circuit as a whole must be offering *less* resistance than either branch alone.

You will not be asked to work out the number. You may be asked to explain exactly this.

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- 1) **State** the potential difference across each branch of a parallel circuit connected to a $6\ \text{V}$ supply. [1]

$6\ \text{V}$ across each branch — every branch gets the full supply p.d.

- 2) **Calculate** the total current when two branches carry $0.30\ \text{A}$ and $0.50\ \text{A}$. [1]

$$I_{\text{total}} = 0.30 + 0.50 = 0.80\ \text{A}$$

- 3) **Calculate** the current in each branch when a $4\ \Omega$ and a $6\ \Omega$ resistor are connected in parallel across a $12\ \text{V}$ supply. [4]

Each branch has the full $12\ \text{V}$. $I_1 = 12/4 = 3.0\ \text{A}$; $I_2 = 12/6 = 2.0\ \text{A}$

- 4) **Calculate** the total current drawn from that supply. [1]

$$I_{\text{total}} = 3.0 + 2.0 = 5.0\ \text{A}$$

- 5) **State** which of those two resistors carries the larger current, and why. [2]

The $4\ \Omega$ resistor. Both branches have the same p.d., so the smaller resistance allows the larger current ($I = V/R$).

- 6) **Explain** qualitatively why adding another resistor in parallel *decreases* the total resistance of a circuit. [2]

Adding a parallel branch opens another route for the charge to flow through. More current is drawn from the supply for the same potential difference, so the circuit as a whole must have a lower total resistance.

- 7) **Explain** what happens to the second lamp when one lamp in a parallel circuit is removed, and why this differs from a series circuit. [3]

The second lamp stays lit. Each branch is a separate route to the supply, so breaking one branch does not break the other. In series there is only one route, so removing any component stops the current everywhere.